

Shot3R: Multi-Shot State Transitions for Streaming 4D Human-Scene Reconstruction

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Figure 1: Shot3R streams successive monocular shots into a common world frame while preserving camera, motion, and identity continuity. Angles denote source-camera rotations from the first shot.

Abstract

Streaming multi-person 4D reconstruction from monocular videos with shot transitions requires camera trajectories, humans, and scene geometry from successive shots to share a common world frame. Existing online recurrent methods update this reconstruction incrementally by carrying a recurrent state forward, an operation that generally assumes visual and geometric continuity between adjacent observations. At a shot boundary, however, the preceding state has conflicting roles: it provides a spatial reference for relating the new shot to the accumulated reconstruction, but becomes unreliable as a recurrent condition within the new shot. Carrying this state forward can corrupt new-shot estimates, whereas resetting it discards the cross-shot spatial relation. **Shot3R** resolves this conflict by formulating each shot transition as a history-conditioned online state transition. At each boundary, Shot3R jointly models the historical representation and the current observation in latent space to infer the spatial relation between the new shot and the existing reconstruction, while initializing an independent recurrent state for the new shot. This boundary-time inference establishes a consistent world-frame representation of camera trajectories, human motion, and scene geometry; geometry-based person association preserves identities across shots, and only the new-shot state is propagated thereafter. In controlled comparisons across three multi-person benchmarks, Shot3R reduces the dataset-balanced mean W-MPJPE by 9.5% and improves mean IDF1 by 0.099 in absolute terms relative to continuous state propagation. On EgoBody’s predefined extreme-viewpoint subset, it reduces W-MPJPE by 29.9%. Comparisons with publicly available online and offline reconstruction systems under the same multi-shot inputs further show competitive reconstruction accuracy.

1 Introduction

Recent advances in streaming 4D reconstruction have made it possible to jointly recover human motion, camera trajectories, and scene geometry from a monocular video as it is processed sequentially (Wang et al., 2025b; Chen et al., 2026b; Sur et al., 2026). These methods achieve efficient online inference by compressing past observations into a recurrent state, but their formulation largely assumes that adjacent frames belong to a continuous capture. Edited videos routinely violate this assumption: they comprise temporally successive shots separated by transitions that may abruptly change viewpoint, remove visual overlap, and alter the visible scene and human appearance (Pavlakos et al., 2022b; Zhang et al., 2025b; On et al., 2026). The history accumulated before such a boundary may therefore no longer provide valid recurrent context for the new shot. We study streaming reconstruction under these discontinuities. Here, *multi-shot* refers to successive segments of a monocular video, rather than synchronized multi-view observations. The goal is to express the camera trajectories, reconstructed humans, and scene geometry of every shot in a *common world frame*—the reconstruction coordinate system initialized by the first shot and shared across the sequence.

Existing methods address only one side of this setting. Multi-shot human reconstruction approaches relate people across transitions by reasoning over multiple available shots, often through joint processing or sequence-level optimization (Pavlakos et al., 2022b; Zhang et al., 2025b; On et al., 2026). Although effective for recovering inter-shot structure, they are generally not designed to produce each estimate from a bounded recurrent state without future-frame access. Stateful online reconstruction methods provide the complementary capability: they process frames in temporal order and avoid per-video global optimization (Wang et al., 2025b; Chen et al., 2026b; 2025). Their recurrent state, however, is designed to evolve through continuous observations and provides no explicit mechanism for a sudden change in history validity at a shot boundary. Neither line of work therefore addresses the central requirement here: preserving the spatial reference established by earlier shots while preventing incompatible history from being propagated through reconstruction of the new shot.

This gap arises from a conflict in how recurrent history is used at a shot boundary. Carrying the state across the boundary preserves the coordinate reference established by previous observations, but forces the new shot to interact with history accumulated under a discontinuous viewpoint; errors introduced at the transition can then persist through subsequent recurrent updates. Reinitializing the state removes this interference and restores coherent recurrence within the new shot, but also erases the reference needed to locate that shot in the common world frame. The key question is therefore not whether history should be retained or discarded, but which of its roles should survive the boundary. Our central insight is that history should survive only as a temporary alignment reference: it may be read to establish the inter-shot relation, but should not be propagated as the recurrent state of the new shot. A controlled comparison using the same reconstruction model supports this distinction: state carry-over produces time-varying camera drift rather than a single fixed offset, whereas independent reconstruction remains coherent within the new shot but loses its relation to the preceding world frame (Section 4.3).

To resolve this conflict, we introduce Shot3R, a streaming reconstruction framework that assigns different lifetimes to the two roles of recurrent history. At each shot boundary, Shot3R temporarily consults the preceding state to infer the inter-shot relation, without retaining the resulting state as memory for subsequent reconstruction. Recurrence within the new shot instead starts from an independently initialized state, which becomes the only state propagated forward. Once the spatial relation is established, geometry-based person association maintains identities across the boundary, and a shared camera–human transform expresses the new-shot reconstruction in the common world frame. Shot3R operates without future-frame access, revision of previously emitted estimates, or per-video optimization. Across three multi-person benchmarks, it reduces the dataset-balanced mean W-MPJPE by 9.5% over continuous state propagation and consistently improves camera accuracy and identity continuity; on EgoBody’s predefined extreme-viewpoint subset, the W-MPJPE reduction reaches 29.9%.

Our contributions are:

- We identify and characterize a conflict in recurrent history at shot boundaries: history is necessary for relating successive shots, yet propagating it into the new shot can introduce persistent reconstruction errors, while discarding it entirely removes the inter-shot spatial reference.

- We introduce Shot3R, a streaming multi-shot reconstruction framework that separates how history is read for alignment from how state is propagated for reconstruction. Shot3R retains pre-boundary history only as a temporary reference and advances the new shot using independently initialized recurrence, while maintaining spatial and identity consistency across the boundary.
- Through controlled same-model comparisons and evaluations on three multi-person benchmarks, we show that this separation improves world-frame human reconstruction, camera accuracy, and identity continuity, with its largest gains under substantial viewpoint changes and without future-frame access or per-video optimization.

2 Related Work

Global human motion reconstruction. Human mesh recovery has evolved from fitting parametric body models to feed-forward regression from monocular images and videos (Kanazawa et al., 2018; Kocabas et al., 2020; Goel et al., 2023). Image-based methods primarily estimate human pose and shape in the camera frame, while temporal models and full-frame multi-person approaches improve motion consistency and reasoning in crowded scenes (Sun et al., 2021b; 2022; 2023; Baradel et al., 2024; Wang et al., 2025c; Sun et al., 2024). When the camera itself moves, however, camera-frame predictions do not directly describe global human motion in the surrounding 3D environment. Existing methods therefore recover human trajectories together with camera motion through motion priors and sequence optimization (Yuan et al., 2022; Ye et al., 2023; Kocabas et al., 2024; Li et al., 2024), joint human-camera modeling (Wang et al., 2024b; Shin et al., 2024; Shen et al., 2024; Yin et al., 2024), or perspective and scale calibration (Patel & Black, 2025; Yang et al., 2025). More recent systems further incorporate humans, cameras, and surrounding scene geometry into a unified reconstruction (Müller et al., 2025; Liu et al., 2026b; Zhang et al., 2025c; Li et al., 2026a; Rojas et al., 2025; Yang et al., 2026; Liu et al., 2026a). Offline methods can exploit global constraints from a complete sequence, but typically require iterative joint optimization of human motion, camera trajectories, and scene geometry over the entire video, incurring substantial latency and computation and limiting scalability to growing sequences. Unified feed-forward methods reduce the cost of per-sequence optimization; continuous video streams additionally require reconstruction to be updated online using only current and past observations, with a bounded history representation and stable per-step computation. The question thus shifts from recovering global relationships from a complete video to maintaining them through an evolving history representation during online inference.

Online reconstruction. To this end, online reconstruction processes video in temporal order and continually updates estimates of cameras, geometry, and people. Classical SLAM, learned dense mapping, and feed-forward point-map models provide the foundations for incremental camera localization, scene mapping, and direct 3D prediction, respectively (Teed & Deng, 2021; Sun et al., 2021a; Wang et al., 2024a; Leroy et al., 2024; Zhang et al., 2025a; Wang et al., 2025a). Recent streaming methods further preserve past observations through spatial memory, progressive registration, or a recurrent state (Wang & Agapito, 2025; Liu et al., 2025; Wang et al., 2025b). For human reconstruction, OnlineHMR studies online recovery in world coordinates (Zhao et al., 2026), while Human3R builds on CUT3R’s recurrent reconstruction to jointly recover cameras, scenes, and multiple people online (Chen et al., 2026b). Building on this line, UniCon3R and GUSH3R extend unified reconstruction through human-scene contact and renderable dynamic representations, respectively (Sur et al., 2026; Abe et al., 2026); other streaming 3D methods study long-term memory, dynamic geometry, and stable recurrent updates (Chen et al., 2025; Wu et al., 2025; Zhuo et al., 2025; Chen et al., 2026a; Zheng et al., 2026; Li et al., 2026b). Existing online methods, however, are primarily designed for continuously captured monocular video and typically rely on stable temporal relations and visual overlap between adjacent frames to carry the reconstruction forward. A shot transition abruptly changes the camera viewpoint and visible content, invalidating this continuity assumption even though reconstructions from different shots must still share a coherent spatial relation. Extending online reconstruction to multi-shot video therefore requires consistency to be maintained across shot transitions.

Multi-shot human reconstruction. Multi-shot human reconstruction aims to recover spatially consistent multi-person 3D motion from monocular videos with shot transitions while preserving identities across shots. Early work separately studied person tracking and re-identification in TV con-

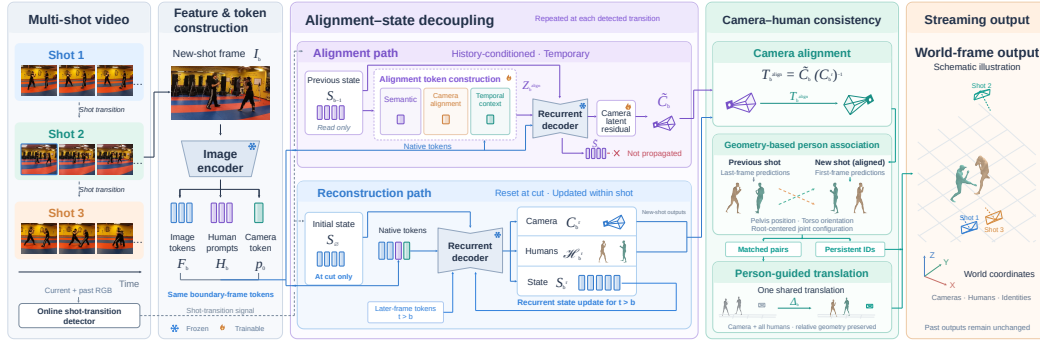


Figure 2: Overview of Shot3R. The history-conditioned path temporarily infers the inter-shot relation, while only the reinitialized path propagates recurrent state. Geometry-based association transfers identities, and a shared transform places new-shot outputs in the common world frame. The rightmost panel is schematic.

tent, single-person recovery across shots, and joint actor–environment reconstruction (Pavlakos et al., 2022a;b;c). HumanMM aligns human orientation and global trajectories across shots through camera-motion estimation and sequence-level motion integration, but focuses on single-person videos (Zhang et al., 2025b). Most closely related to our setting, Multi-THuMBS uses a 3D scene prior to reconstruct observations on both sides of a shot boundary in a common space, associates people using appearance, pose, and geometric cues, and then optimizes the global result with trajectory smoothing and a cross-camera reprojection constraint (On et al., 2026). It extends multi-shot reconstruction to multiple people, but adopts multi-stage global optimization over an observed sequence. ShowMak3R also handles multi-person, multi-shot content, but optimizes per-video neural representations of actors and scenes, targeting renderable cinematic reconstruction rather than online human-motion recovery (Kim et al., 2025). Cross-shot identity persistence is further related to 3D-aware tracking and appearance-, pose-, or keypoint-based re-identification (Rajasegaran et al., 2022; He et al., 2021; Yuan et al., 2025; Somers et al., 2024). Together, these works advance multi-shot human reconstruction through global motion recovery, spatial alignment, and person association, but commonly jointly process observations across the shot boundary or optimize a complete sequence. They therefore leave open how a maintained history representation should cross a shot boundary during online recurrent reconstruction. Shot3R addresses this streaming setting by using history temporarily to establish the inter-shot spatial relation, without propagating it as the recurrent state of the new shot, thereby maintaining spatial and identity consistency under sequential multi-shot processing.

3 Method

Shot3R processes a multi-shot monocular video in temporal order and reconstructs its cameras, visible people, and scene geometry in a common world frame. Within each shot, a recurrent reconstructor continually updates a compact history representation. At a shot transition, this history remains a useful reference for establishing the inter-shot spatial relation, but is no longer a reliable recurrent condition for the new visual content. Shot3R therefore assigns different information lifetimes to these two roles of history.

As illustrated in Figure 2, a detected boundary triggers two complementary evaluations of the same first new-shot frame. A history-conditioned alignment path temporarily reads the preceding state to infer the inter-shot relation and is then discarded. A reinitialized reconstruction path starts the new shot independently and owns the only state propagated forward. Typed alignment representations organize boundary evidence from the current observation and recurrent history. The resulting relation, geometry-based person association, and a shared transform then commit the reinitialized outputs to the existing world frame. This boundary operation uses only current and past observations, revises no previously emitted frame, and requires no per-video optimization.

3.1 Recurrent Reconstruction at Shot Boundaries

Given a monocular stream $\{I_t\}_{t=1}^T$ composed of consecutive video shots, Shot3R uses only observations up to time t to recover camera pose C_t , scene geometry P_t^w , and multiple visible people $\mathcal{H}_t$ online. All outputs are expressed in a *common world frame*: the reconstruction coordinate system initialized by the first shot and shared by the cameras, people, and scene geometry of every subsequent shot, rather than an externally specified absolute coordinate system. Encoding past observations in a recurrent state S_t , a general stateful reconstruction process is

$$(y_t, S_t) = f_\theta(I_t, S_{t-1}), \quad y_t = \left(C_t, P_t^w, \{J_t^i, V_t^i, q_t^i\}_{i=1}^{N_t}\right), \quad (1)$$

where C_t is a camera-to-world pose, and J_t^i , V_t^i , and q_t^i denote the SMPL-X (Pavlakos et al., 2019) joints, mesh vertices, and anonymous slot of person i , respectively. We instantiate f_θ with a pre-trained Human3R recurrent reconstructor, which supplies within-shot camera, human, and scene predictions (Chen et al., 2026b). Under continuous observations, the spatial-reference and recurrent-conditioning roles of S_{t-1} are compatible. A shot transition separates them: the preceding state still relates the new observation to the established reconstruction, but its viewpoint- and content-specific representation is no longer a valid condition for updating the new-shot state. Directly carrying it forward preserves the reference but can propagate this mismatch; reinitializing it isolates the mismatch but removes the inter-shot relation.

To resolve this conflict, Shot3R performs two complementary online evaluations at a detected boundary b :

$$\begin{aligned} (\tilde{y}_b, \tilde{S}_b) &= f_{\theta, \phi}(I_b, S_{b-1}), & (y_b^r, S_b^r) &= f_\theta(I_b, S_\emptyset), \\ (y_t^r, S_t^r) &= f_\theta(I_t, S_{t-1}^r), & t &> b. \end{aligned} \quad (2)$$

The history-conditioned alignment path reads S_{b-1} only to infer the relation between the new shot and the existing world frame; its predictions and $\tilde{S}_b$ are discarded after the boundary operation. The reinitialized reconstruction path starts from $S_\emptyset$, and S_b^r becomes the sole recurrent state for all subsequent frames in the shot. Thus, inter-shot relation estimation and new-shot state propagation have separate information paths.

3.2 History-Conditioned Shot Alignment

The history-conditioned path must estimate an inter-shot relation from the interaction between the first new-shot frame I_b and the preceding state S_{b-1} . Recurrent features designed for continuous reconstruction mix image content, camera motion, human information, and historical context, whereas boundary reasoning benefits from keeping their different roles identifiable. Rather than prematurely collapsing them into a single correction vector, Shot3R explicitly constructs three typed alignment representations for semantic correspondence, camera-latent change, and accumulated temporal context.

Let c_b denote observation evidence pooled from the current image, camera, and human tokens, and let m_{b-1} denote historical evidence retrieved from the recurrent state and camera memory. The typed representations are

$$\begin{aligned} z_b^{\text{sem}} &= \alpha_b c_b + (1 - \alpha_b) m_{b-1}, & z_b^{\text{cam}} &= \text{MLP}_{\text{cam}}[p_b, p_{b-1}, \Delta p_b, m_{b-1}], \\ z_b^{\text{temp}} &= \text{MLP}_{\text{temp}}[\bar{z}_{b-1}, \delta_{b-1}, g_{b-1}], & z_b^{\text{align}} &= [z_b^{\text{sem}}, z_b^{\text{cam}}, z_b^{\text{temp}}]. \end{aligned} \quad (3)$$

Here α_b adaptively balances current semantic evidence and remembered context; p_b and p_{b-1} are camera tokens on the two sides of the boundary, with $\Delta p_b = p_b - p_{b-1}$; and $\bar{z}_{b-1}$, δ_{b-1} , and g_{b-1} denote the preceding alignment summary, camera-latent residual, and residual gate. These are not independent reconstruction modules. They organize heterogeneous boundary evidence so that its source remains identifiable when entering the decoder.

After adding type embeddings, Z_b^{align} is inserted after the native camera token and decoded jointly with the image, camera, and human representations by the frozen recurrent backbone. The decoded boundary representations produce gated camera and human latent residuals. The former updates the camera representation before the original pose head predicts the temporary camera $\tilde{C}_b$; the latter incorporates person-translation constraints into the same boundary inference. All typed representations and latent updates exist only in the alignment path and are never written into the state propagated through the new shot. Pooling and residual details are given in Appendix C.

We train the alignment representations with both transition and continuous four-frame sequences. Transition samples follow $A(t), A(t+1), B(t+2), B(t+3)$ for two cameras A and B , supervising changes across viewpoint transitions; continuous samples contain four successive frames from one camera and preserve the original behavior under continuous input.

The alignment objective is

$$\mathcal{L}_{\text{align}} = \mathcal{L}_t + 5\mathcal{L}_R + 10\mathcal{L}_h + 0.05(\mathcal{L}_g + \mathcal{L}_{\text{imp}}) + 10^{-5}(\mathcal{R}_{\text{cam}} + \mathcal{R}_{\text{human}}), \quad (4)$$

where $\mathcal{L}_t$ and $\mathcal{L}_h$ are Smooth-L1 camera and human translation losses, and $\mathcal{L}_R$ is quaternion geodesic distance. With e and e^{raw} denoting normalized corrected and unmodified camera errors, respectively, the gate and improvement terms are $\mathcal{L}_g = \text{BCE}(g, \text{clip}(e^{\text{raw}}, 0, 1))$ and $\mathcal{L}_{\text{imp}} = [e - e^{\text{raw}}]_+$. The remaining terms regularize camera and human latent updates. Because a transition changes the relation between history and the current observation rather than the within-shot reconstruction target, we optimize only the typed representations, camera and human latent-residual branches, and low-rank prediction-head adapters (Hu et al., 2022). The image encoder, recurrent backbone, base heads, and Multi-HMR component (Baradel et al., 2024) remain frozen.

3.3 Cross-Shot Spatial and Identity Consistency

The reinitialized path isolates new-shot recurrence from incompatible history, but its predictions inhabit an independently initialized local frame. An inter-shot relation is useful only if it can commit these predictions to the common world frame without reintroducing the preceding recurrent state. Shot3R therefore retains only the camera relation inferred at the boundary and applies it to the reinitialized outputs; the temporary alignment state has no path into later recurrence. Since the two paths process the same image, their relative camera pose captures the relation induced by historical conditioning. Let C_b^r be the camera-to-world pose from the reinitialized path, and let $\tilde{C}_b$ be the temporary alignment-path pose expressed in the accumulated common world frame. The initial mapping of the new-shot local frame is

$$T_b^{\text{align}} = \tilde{C}_b(C_b^r)^{-1}. \quad (5)$$

It satisfies $T_b^{\text{align}}C_b^r = \tilde{C}_b$ and therefore maps the reinitialized camera and human predictions into the existing frame. For later transitions, $\tilde{C}_b$ includes the previously accumulated inter-shot mapping. Importantly, T_b^{align} results from feature interaction between the current boundary observation and recurrent history, rather than post-hoc registration of two fully reconstructed shots.

Camera alignment alone does not determine person correspondence. After the initial spatial mapping, Shot3R associates people one-to-one using their predicted root locations, torso orientations, and root-centered joint configurations. For cue $a \in \{\text{root}, \text{torso}, \text{joint}\}$, we normalize its pairwise cost matrix and form the combined cost as

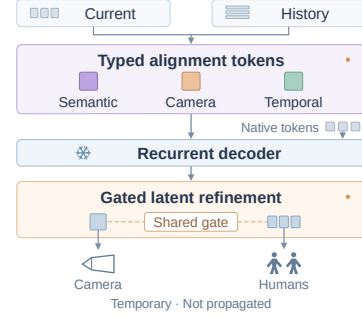


Figure 3: **History-conditioned shot alignment.** Typed representations organize current and historical evidence; gated latent updates remain temporary.

$$\widehat{d}_{ij}^a = \frac{d_{ij}^a}{\max(\text{median}_{u,v} d_{uv}^a, \epsilon)}, \quad d_{ij} = \widehat{d}_{ij}^{\text{root}} + \widehat{d}_{ij}^{\text{torso}} + \widehat{d}_{ij}^{\text{joint}}. \quad (6)$$

Here the root term is Euclidean distance, the torso term is angular distance, and the joint term averages Euclidean distances between root-centered joints. Hungarian assignment (Munkres, 1957) gives the matched set $\mathcal{M}_b$. Matching uses no identity labels, external camera calibration, or future frames. Matched people inherit their preceding anonymous identities, whereas unmatched new-shot slots receive new identities. This permutation is established once at the first new-shot frame; subsequent frames retain the native slots maintained by the reinitialized recurrence.

The initial camera mapping establishes orientation and the primary world relation, while matched people provide a direct constraint on the remaining shared translation. Let r_{b-1}^i be the predicted pelvis center of person i in the final preceding frame and $\bar{r}_b^j$ the pelvis center of person j after the initial mapping. We estimate

$$\Delta_b = \lambda \text{median}_{(i,j) \in \mathcal{M}_b} (r_{b-1}^i - \bar{r}_b^j), \quad \lambda = 0.5, \quad (7)$$

where the median is taken component-wise over matched pelvis displacements and the fixed λ is unchanged across datasets. If no valid match is available, the translation is omitted and only the camera-derived mapping is retained. With $T_b^\Delta = \begin{bmatrix} I_3 & \Delta_b \\ \mathbf{0}^\top & 1 \end{bmatrix}$ and $T_b = T_b^\Delta T_b^{\text{align}}$, the committed outputs for $t \geq b$ are

$$\begin{aligned} C'_t &= T_b C_t^r, & J_t'^i &= T_b J_t^{r,i}, & V_t'^i &= T_b V_t^{r,i}, \\ P_t'^w &= C'_t P_t^{r,c} = T_b P_t^{r,w}, & q_t'^i &= \pi_b(q_t^{r,i}), & S'_t &= S_t^r. \end{aligned} \quad (8)$$

Here π_b is the boundary identity mapping, and 3D points are understood in homogeneous coordinates. The camera-local scene point map $P_t^{r,c}$ is left unchanged; its world-space realization follows from the transformed camera pose. Applying one shared world transform to cameras and all people preserves their relative geometry, while retaining S_t^r ensures that only the reinitialized state survives into future reconstruction. The transform is cached and applied as subsequent frames are processed. It uses no future frame, performs no sequence-level optimization, and leaves all previously emitted outputs unchanged.

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